

ECON 300 — Midterm 1

Detailed Solution Key

Note: Diagram questions reference the figures provided in the exam PDF (pages 10–12).

Question 1 *(Exam PDF p. 1)*

Correct answer: A

In the neoclassical model, a higher wage makes leisure more expensive (substitution effect → more work), but also raises purchasing power (income effect → more leisure if leisure is normal). Empirically, these effects often partially offset, so the **total** (uncompensated) wage elasticity of labour supply is usually positive but small in magnitude.

Textbook reference: Ch. 2: Labour supply elasticities; substitution vs income effects.

Question 2 *(Exam PDF p. 1)*

Correct answer: B

If the worker is **constrained** to work fewer hours than she would choose, she cannot reach the tangency optimum (where an indifference curve is tangent to the budget line). Any binding hours constraint forces her to a feasible point that yields **lower utility** than the unconstrained tangency point, so she must lie on a lower indifference curve.

Textbook reference: Ch. 2: Utility maximization with constraints; tangency vs constrained outcomes.

Question 3 *(Exam PDF p. 1)*

Correct answer: C

The reservation wage is the minimum wage that makes the worker willing to participate (work) rather than choose non-work. If the offered wage is below the reservation wage, the worker supplies zero hours.

Textbook reference: Ch. 2: Participation decision and reservation wage.

Question 4 (Exam PDF p. 1)**Correct answer: C**

At a zero-hours (corner) solution, the worker chooses all leisure and no work. This occurs when the indifference curve at that point is **steeper** than the budget line: the worker values an extra hour of leisure more than the market compensates ($|MRS| > w$). Therefore, the statement that the indifference curve is **flatter** than the budget line is false.

Textbook reference: Ch. 2: Corner solutions; MRS vs wage.

Question 5 (Exam PDF p. 2)**Correct answer: A**

In the income–leisure model, the slope of the budget line is determined by the market wage: each hour less leisure (more work) increases income by w . Non-labour income shifts the budget line up/down (intercept), but it does not determine the slope.

Textbook reference: Ch. 2: Budget constraint slope/intercept.

Question 6 (Exam PDF p. 2)**Correct answer: D**

Holding utility constant, a wage increase makes leisure relatively more expensive, so the substitution effect always pushes toward **less leisure** and **more hours worked**. The overall effect can still be ambiguous because the income effect may offset it.

Textbook reference: Ch. 2: Substitution effect from a wage change.

Question 7 (Exam PDF p. 2)**Correct answer: D**

At an interior optimum, MRS equals the wage, and the budget line is tangent to the indifference curve. However, there is no general reason for the income effect to equal the substitution effect; they depend on preferences and the magnitude of the wage change.

Textbook reference: Ch. 2: Interior optimum conditions; decomposition of wage effects.

Question 8 *(Exam PDF p. 2)*

Correct answer: A

Labour force participation is the decision to allocate time to labour-market activity (working or actively searching) versus non-market activity. It is not restricted only to being currently employed.

Textbook reference: Ch. 2: Participation and labour force definitions.

Question 9 *(Exam PDF p. 3)*

Correct answer: C

In standard labour force statistics, the unemployed are (i) without a job and (ii) actively seeking work (and available). Merely being jobless is not sufficient if a person is not searching.

Textbook reference: Ch. 2: Unemployment definition and measurement.

Question 10 *(Exam PDF p. 3)*

Correct answer: A

Labour force surveys typically sample the civilian, non-institutional population (excluding those in institutions and, depending on the definition, certain groups such as the armed forces).

Textbook reference: Ch. 2: Labour force survey population.

Question 11 *(Exam PDF p. 3)*

Correct answer: C

A demogrant is a lump-sum transfer (often universal within a demographic group). It does not vary with the worker's wage. Because it is lump-sum, it shifts the budget constraint upward without changing its slope.

Textbook reference: Ch. 3: Demogrant and income maintenance.

Question 12 (Exam PDF p. 3)

Correct answer: B

A demogrant shifts income up at every leisure level but does not change the trade-off between leisure and income (slope unchanged). That implies an income effect only (if leisure is normal, recipients choose more leisure / fewer hours).

Textbook reference: Ch. 3: Income effect of lump-sum transfers.

Question 13 (Exam PDF p. 4)

Correct answer: B

“Negative” income tax means that at low earnings the individual receives a net payment (a credit/subsidy) instead of paying positive taxes. As earnings rise, the transfer is phased out via a tax-back rate.

Textbook reference: Ch. 3: Negative income tax structure.

Question 14 (Exam PDF p. 4)

Correct answer: A

Unlike conventional welfare with very high implicit tax rates (sometimes near 100%), a negative income tax allows recipients to keep part of additional earnings because benefits are clawed back only partially (at rate $t < 1$).

Textbook reference: Ch. 3: Welfare vs NIT; implicit tax rates.

Question 15 (Exam PDF p. 4)

Correct answer: E

The diagram shows a guaranteed income level G (vertical intercept) and a benefit phase-out region where the slope is flatter than the market-wage slope (because benefits are clawed back). That pattern corresponds to a Negative Income Tax: $C = G + (1-t)wh$ up to the break-even point, then $C = wh$ after benefits exhaust.

Textbook reference: Ch. 3: Budget constraints under NIT; guaranteed income and tax-back rate.

Question 16 (*Exam PDF p. 4*)

Correct answer: D

Canada's EI replaces a constant fraction of insurable earnings (roughly 55%) up to a maximum insurable earnings cap; above the cap, benefits do not increase further.

Textbook reference: Ch. 3: Employment Insurance (replacement rate and cap).

Question 17 (*Exam PDF p. 5*)

Correct answer: A

Disability insurance can reduce work because it provides income without work, raising non-labour income and lowering the need to work. This generates a (potentially strong) income effect toward more leisure; program rules can also reduce the net return to work in some ranges.

Textbook reference: Ch. 3: Disability insurance and work incentives.

Question 18 (*Exam PDF p. 5*)

Correct answer: D

A classic feature discussed for Canada's EI system is relatively high generosity/availability for seasonal industries, which can interact with seasonal patterns of employment.

Textbook reference: Ch. 3: EI and seasonal unemployment.

Question 19 (*Exam PDF p. 5*)

Correct answer: B

Childcare subsidies reduce the fixed costs of working (e.g., out-of-pocket childcare costs). Lower fixed costs increase the attractiveness of labour force participation, especially for secondary earners.

Textbook reference: Ch. 3: Fixed costs of work; childcare policies and participation.

Question 20 (Exam PDF p. 5)

Correct answer: A

If a refundable benefit makes working pay more at the margin, it raises the net wage (or lowers the implicit tax rate), generating a substitution effect toward more hours worked (leisure becomes more expensive).

Textbook reference: Ch. 2 & Ch. 3: Substitution effect; refundable credits.

Question 21 (Exam PDF p. 6)

Correct answer: D

Use elasticity: $\varepsilon = (\Delta L/L)/(\Delta w/w)$.

Given $L = 100$ hours, w rises from 10 to 12, so $\Delta w/w = 2/10 = 0.20$ (20%).

$$\Delta L/L = \varepsilon \cdot (\Delta w/w) = (-1.2)(0.20) = -0.24.$$

So labour falls by 24% of 100 = 24 hours.

Textbook reference: Ch. 5: Elasticity of labour demand; percentage-change calculations.

Question 22 (Exam PDF p. 6)

Correct answer: E

Across industries, lower pay is closely tied to lower skill requirements, lower marginal product, and weaker bargaining power. Among the options given, low unionization best captures weaker bargaining/collective wage effects in fast-food employment.

Textbook reference: Ch. 5: Wage determination and bargaining institutions (plus productivity/skill discussion).

Question 23 (Exam PDF p. 6)

Correct answer: B

A monopolist faces downward-sloping product demand, so marginal revenue (MR) falls with output and is below price ($MR < P$). Labour demand is based on marginal revenue product (MRP), which is lower and typically steeper (less elastic) than when output price is constant under competition.

Textbook reference: Ch. 5: Labour demand under imperfect competition; MRP vs VMP.

Question 24 (Exam PDF p. 6)

Correct answer: E

Elasticity -0.55 means a 1% wage increase reduces labour demanded by 0.55% (in absolute value). That is the definition of elasticity in percentage terms.

Textbook reference: Ch. 5: Interpreting labour-demand elasticities.

Question 25 (Exam PDF p. 6)

Correct answer: E

Cutting school funding shifts the demand for teachers left (lower willingness/ability to hire), so teacher demand would fall, not rise. Therefore the statement claiming demand rises when funding is cut is false.

Textbook reference: Ch. 5: Demand shifts from budget/product demand changes.

Question 26 (Exam PDF p. 7)

Correct answer: B

In the long run all inputs are variable. Firms can adjust capital, technology choice, and organizational form, so both labour and capital can change.

Textbook reference: Ch. 5: Short run vs long run in labour demand.

Question 27 (Exam PDF p. 7)

Correct answer: A

A profit-maximizing competitive firm hires labour until the value of marginal product equals the wage: $VMPL = w$ (or $MRP = w$). The marginal product alone is measured in physical units and must be valued at output price to compare with w .

Textbook reference: Ch. 5: Hiring rule $VMPL = w$.

Question 28 (Exam PDF p. 7)

Correct answer: C

For a competitive firm, labour demand is the downward-sloping portion of the VMPL curve: the firm hires up to the point where VMPL intersects the wage. The upward portion would never be chosen.

Textbook reference: Ch. 5: Labour demand as VMPL (downward-sloping segment).

Question 29 (Exam PDF p. 7)

Correct answer: D

Cost minimization for a given output occurs where the isoquant is tangent to the *lowest* isocost line that can produce that output (not the highest isoquant).

Textbook reference: Ch. 5: Isoquants, isocosts, and cost minimization.

Question 30 (Exam PDF p. 7)

Correct answer: D

Long-run input choice satisfies $MRTS = w/r$ (or $MPPL/MPPK = w/r$), which describes cost-minimizing input mix. The firm does not maximize “profit per worker”; it maximizes total profit.

Textbook reference: Ch. 5: Long-run cost minimization and profit maximization.

Question 31 (Exam PDF p. 8)

Correct answer: D

Scale effect: when wages rise, marginal cost increases, output falls, and the firm reduces all variable inputs, including labour (movement to a lower isoquant).

Textbook reference: Ch. 5: Scale effect of wage increases.

Question 32 *(Exam PDF p. 8)*

Correct answer: C

The slope of an isoquant at a point is the marginal rate of technical substitution (MRTS), describing how much capital can replace labour while holding output constant.

Textbook reference: Ch. 5: MRTS and isoquants.

Question 33 *(Exam PDF p. 8)*

Correct answer: A

An isocost holds total cost constant given factor prices; along a single isocost, factor prices are fixed. (Quantities can vary while keeping C constant.)

Textbook reference: Ch. 5: Isocost lines.

Question 34 *(Exam PDF p. 8)*

Correct answer: A

If domestic and foreign labour are complements, substitution away from domestic labour is limited, but the wage increase raises costs and reduces output (scale effect), which lowers demand for both domestic and foreign labour. Net effect is away from domestic labour.

Textbook reference: Ch. 5: Complements/substitutes; substitution vs scale effects.

Question 35 *(Exam PDF p. 8)*

Correct answer: E

The scale effect is obtained by tracking the change in output (movement to a different isoquant) after the wage change, not by the within-isoquant substitution response.

Textbook reference: Ch. 5: Decomposing total effect into substitution and scale.

Question 36 *(Exam PDF p. 9)*

Correct answer: B

If substitute inputs are hard to expand (inelastic supply), firms cannot easily substitute away from labour when wages rise, making labour demand more inelastic.

Textbook reference: Ch. 5: Determinants of labour-demand elasticity (substitutability and input supply).

Question 37 (*Exam PDF p. 9*)

Correct answer: D

If Canadian and foreign labour are complements, cheaper foreign labour can expand production (scale effect) and increase demand for both foreign and Canadian labour.

Textbook reference: Ch. 5: Globalization/outsourcing with complements.

Question 38 (*Exam PDF p. 10*)

Correct answer: A

Increasing the offered wage steepens the post-benefit segment of the budget constraint (higher return to work) and rotates the constraint accordingly. The top-left panel shows this wage increase change in slope.

Textbook reference: Ch. 3: Budget constraints and work incentives; wage increases.

Question 39 (*Exam PDF p. 10*)

Correct answer: B

Reducing the implicit tax rate (lower t) makes the phase-out region less flat (net wage rises), rotating the benefit region upward. The top-right panel corresponds to lowering the tax-back rate.

Textbook reference: Ch. 3: Tax-back rate and net wage in the phase-out region.

Question 40 (*Exam PDF p. 10*)

Correct answer: D

A preference shift toward valuing work (or lower disutility of work) is represented by indifference curves changing so that chosen hours rise holding the constraint fixed. The bottom-right panel shows this preference shift.

Textbook reference: Ch. 2/3: Preferences and labour-leisure choice; policy environment fixed.

Question 41 (Exam PDF p. 11)

Correct answer: A

EITC has a phased-in region (increasing net wage), a plateau, and a phase-out region. Diagram (a) shows this characteristic piecewise structure.

Textbook reference: Ch. 3: Earned Income Tax Credit budget constraint.

Question 42 (Exam PDF p. 11)

Correct answer: B

A wage subsidy increases the effective wage (steeper budget line) over a range of hours/earnings. Diagram (b) shows a higher slope consistent with subsidizing earnings.

Textbook reference: Ch. 3: Wage subsidy and increased net wage.

Question 43 (Exam PDF p. 11)

Correct answer: C

Negative income tax shows guaranteed income at zero earnings plus a partial clawback (flatter slope while on benefits) until a break-even point. Diagram (c) matches that structure.

Textbook reference: Ch. 3: Negative income tax (G and tax-back).

Question 44 (Exam PDF p. 12)

Correct answer: A

With wage rising from W_0 to W_1 (r fixed), the substitution effect is the movement along the original isoquant to the tangency with the new relative prices: $N_0 \rightarrow NS$.

Textbook reference: Ch. 5: Substitution effect in the isoquant–isocost decomposition.

Question 45 (Exam PDF p. 12)

Correct answer: C

The scale effect is the additional movement due to reduced output after the wage increase (moving to a lower isoquant): $NS \rightarrow N_1$.

Textbook reference: Ch. 5: Scale effect after substitution effect.